

Equipment accountability. One tap at a time.

TRAKR / Native iOS + Android / School equipment checkout

Know who borrowed it.
Know what came back.

A complete handover record for shared school equipment.

TAP → RECORD

ITEM
BORROWER
CONDITION
RETURN

5-minute pitch / 3-minute live demonstration

The handover is where context gets lost.

Illustrative school scenario: a student borrows a camera and tripod.

01

Who has it?

Borrower details can be incomplete or scattered.

02

Was it damaged?

Condition needs to be recorded when custody changes.

03

Did it return?

A return needs a clear closing action and history.

Opportunity: make accountability part of the handover itself.

One workflow. Three people who benefit.

Start with one equipment room and one teacher champion.

01

Student

Scan several items.
Acknowledge condition. Keep
a receipt.

02

Teacher

Enroll tags. View claims.
Confirm returns. Resolve
issues.

03

Equipment lead

Review inventory, overdue
claims and operational
exceptions.

User value: less ambiguity for students; clearer follow-up for staff.

Tap. Check. Confirm. Return.

Example: a camera with a loose strap and a tripod with no issues.

01 Tap

Resolve the NFC tag to an enrolled item.



02 Check

Review the list and report condition.



03 Confirm

Save claims and show a receipt.



04 Return

Teacher closes active claims.

Condition reporting and return confirmation are separate responsibilities.

A tap identifies the item. Rules govern the write.

Mobile clients write directly to Firestore using atomic batches.



Google sign-in → verified identity | Teacher return → claim status update

Atomic commit: all related writes succeed together, or none do.

Designed for the awkward moments, too.

The quality of the system shows up when a workflow goes wrong.

01

Role boundaries

Verified school identity and record ownership checks.

02

Honest failure

Keep staged items offline; require a connection to confirm.

03

Durable history

Replace a tag without replacing the equipment record.

Repository tests exist. Hardware and school-environment proof still matter.

Measure the handover, not just the download.

Proposed pilot: one equipment room, four weeks, a teacher sponsor.

01

Week 1

Baseline timing, teacher interviews and tag enrollment.

02

Weeks 2-3

Observe live handovers, failures and support requests.

03

Week 4

Compare results and decide whether to expand.

Targets: $\geq 30\%$ faster median checkout • $\geq 95\%$ complete records

Start narrow. Earn the right to expand.

Business hypothesis: a school subscription with supported onboarding.

01

Buyer

School or department,
supported by a teacher
champion.

02

Revenue hypothesis

Annual subscription; optional
tag and setup support.

03

Expansion

One room → more
departments → additional
schools.

Next proof: repeat usage, willingness to pay and support cost per school.

Give us one room to prove the value.

A focused pilot turns a working MVP into evidence.

1 room. 1 teacher. 4 weeks.

Deliverable: measured time savings, record quality and a clear go / no-go decision.

NEXT / Watch one student checkout become a teacher-confirmed return.

Three minutes. One complete handover.

Show the record changing, not every menu in the app.

0:00-1:00

Student + two items
Scan into staging



1:00-1:35

Report condition
Confirm + receipt



1:35-2:20

Teacher reviews
Confirm return



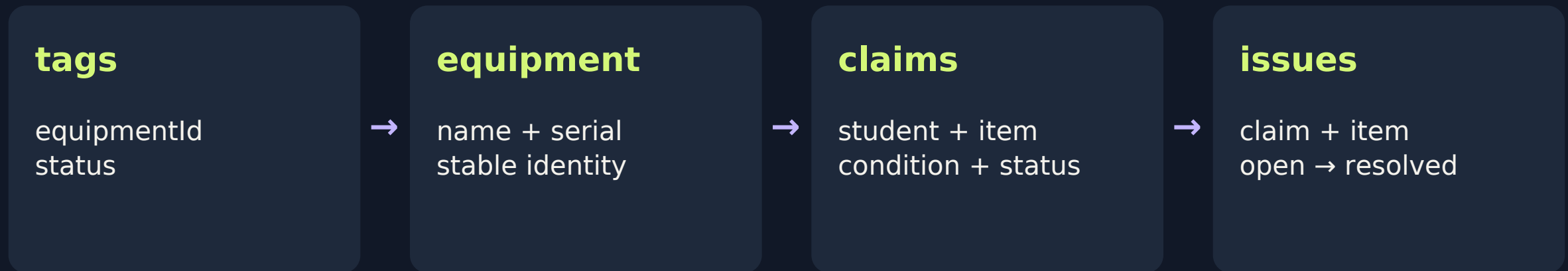
2:20-3:00

History + issue
Explain next action

Success: the audience sees who borrowed it, its condition and its return.

Identity stays stable when tags change.

Simplified data relationships; names follow the Firebase implementation.



checkoutBatches → claims ← returnBatches

Tag replacement preserves history. Multiple active claims are currently allowed.

Two access paths. Two enforcement points.

Mobile database access and privileged operations access must both be secured.



An MVP with clear limits.

These constraints define the next engineering work.

01

Connectivity

Confirmation requires internet. Overdue checks run in-app.

02

Identity & custody

School-specific roles. NFC tags are identifiers, not anti-theft devices.

03

Launch readiness

Validate hardware, school OAuth, concurrency and demo isolation.

Roadmap: push delivery, tenant isolation and stronger operational validation.

Make each claim testable.

A proposed assessment map, since no course rubric was supplied.

Problem understanding

User scenario + teacher interviews to collect

Design & implementation

Journey + data flow + live end-to-end demo

Testing & reliability

Rules/store/UI tests in repository; device checks pending

Critical evaluation

Constraints, alternatives and pilot acceptance criteria

Communication

Five-minute argument + three-minute demonstration

Why this workflow instead of another log?

Conceptual comparison; no vendor benchmarking has been performed.

01

Paper / sheets

Low setup burden. Handover completeness depends on process discipline.

02

QR / NFC forms

A credible lightweight option. Compare scan reliability and completion.

03

Trakr focus

A connected checkout, condition, receipt and teacher-return workflow.

Differentiation hypothesis: workflow completion, not the tag technology.

Set the decision rule before the pilot.

Record baseline and pilot samples; report sample size and failures.

Median checkout time

Target: $\geq 30\%$ below the observed baseline

Record completeness

Target: $\geq 95\%$ of eligible handovers complete

Student data isolation

Target: 0 unauthorized reads in the test set

Workflow reliability

Track failed commits, duplicates and recovery

Adoption & follow-up

Measure weekly use and staff minutes spent chasing

Prove willingness to pay before scaling.

Early business exploration; there is no validated revenue forecast.

$$2 \text{ hours} \times \$30 \times 40 \text{ weeks} = \$2,400$$

Illustrative annual time value per school — not measured savings or revenue.

01

Discover

Who owns the budget? What evidence unlocks a purchase?

02

Cost

Tags, setup, hosting, support and acquisition all count.

03

Gate expansion

Paid conversion, retention, tenant isolation and commercial rights.

A pitch grounded in the repository.

Implementation evidence is not the same as field validation.

Product scope

README.md · docs/production-integration.md

Data & mutation flow

Trakr/Models/Domain.swift · Services/FirebaseGateway.swift

Security & operations

firebase/firestore.rules · dashboard/README.md

Verification coverage

docs/mvp-verification.md · TrakrTests · firebase/functions/test

Next evidence

Teacher interviews · device results · pilot data · buyer feedback